

PHYS 427: Lecture # 2

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1. Multiplicity of Physical Systems

Review

Useful:

$$(p+q)^N = \sum_{n=0}^N \binom{N}{n} p^n q^{N-n} \quad (1)$$

One can also take the following derivative:

$$p \partial_p p^n = n p^n \quad (2)$$

Then, taking the derivative of both sides of Eq.(1) and multiply it by p , then we obtain:

$$Np(p+q)^{N-1} = \sum_{n=0}^N \binom{N}{n} n p^n q^{N-n} \quad (3)$$

$$Np = \sum_{n=0}^N \binom{N}{n} n p^n q^{N-n} \quad (4)$$

Therefore, $\langle n \rangle = pN$. Such a procedure can be repeated for $\langle n^2 \rangle$. Here,

$$\langle n^2 \rangle = \sum n^2 \binom{N}{n} p^n q^{N-n} \quad (5)$$

Moreover,

$$(p \partial_p)^2 p^n = n^2 p^n \quad (6)$$

Then,

$$\sum \binom{N}{n} n^2 p^n q^{N-n} = (p \partial_p)^2 (p+q)^N \quad (7)$$

$$= p \partial_p Np(p+q)^{N-1} \quad (8)$$

$$= Np(p+q)^{N-1} + N(N-1)p^2(p+q)^{N-2} \quad (9)$$

$$\langle n^2 \rangle = Np + N(N-1)p^2 \quad (10)$$

Here, $P(n)$ is a peak. The width of that peak is shown as

$$\sigma_n^2 = \langle (n - \langle n \rangle)^2 \rangle \quad (11)$$

$$= \langle n^2 \rangle - \langle n \rangle^2 \quad (12)$$

The relative width can be calculated as $\sigma_n/\langle n \rangle$, and is useful for comparing the true standard deviation versus the magnitude of the atoms we are dealing with. E.g., a $\sigma_n \sim 1$ and $\mu \sim 10^{23}$ means that σ_n doesn't really mean anything. This is essentially a delta function.

Multiplicity of a physical system

We assume that the system is closed (i.e., an isolated system). This also means that the system does not interact with the outside environment. Then, let U be the total energy of the system. Then,

U is constant in the isolated system

Later, U can exchange with some sort of bath (environment). The macrostate of the system is defined as $\Omega(U, N)$ where N is the total number of atoms.

Einstein model of solid

Assume that we have a 2d lattice of atoms, and each atom are coupled to each other by springs. Then, treat all the atoms as a collection of N , independent quantum harmonic oscillators. As a reminder, the energy levels are

$$E_i = \hbar\omega(n_i + 1/2), \quad n_i = 0, 1, 2, \dots \quad (13)$$

Here, i is the index of each atom. Then,

$$U = \sum_i E_i = \hbar\omega(n + \frac{N}{2}) \quad (14)$$

Here, we neglect the zero point energy since $\hbar\omega N/2$. Moreover, we also have that:

$$n = \sum_i s_i$$

So then, what is $\Omega(U, N)$? In other words, how many microstates correspond to the same total energy U correspond to n quanta? This is possible when

$$s_1 + s_2 + \dots = n \quad (15)$$

This is also the integer partitions of n . We can imagine drawing out n different dots (up to n quanta) and dividing them up. There is a maximum of $N - 1$ partitions.

Then, there are a total of $n + (N - 1)$ slots. Then, the binomial coefficient is

$$\binom{n + (N - 1)}{n} \quad (16)$$

This is also the same as

$$\Omega(n, N) = \frac{(n + N - 1)!}{n!(N - 1)!} \quad (17)$$

This is same as choosing the arrangements of n quanta and $N - 1$ partitions.

Here, n is the number of quanta and $N - 1$ is the number of partitions between the atoms. Therefore, there is a total number of $n + N - 1$ arrangements of the partitions and the quanta. We also choose n quanta, which is to say how many times are we able to arrange n in these slots given that the partitions remain "still".

We can also approximate

$$\Omega(n, N) = \frac{(n + N - 1)!}{n!(N - 1)!} \quad (18)$$

Using stirling's approximation for $N \gg 1$, we derive that

$$\ln(N!) = \ln [N(N-1)(N-2)\dots] \quad (19)$$

$$= \ln N + \ln(N-1) + \dots \quad (20)$$

$$= \sum_x \ln x \approx \int_1^N \ln(x) dx \quad (21)$$

$$\approx N \ln N - N \quad (22)$$

In an Einstein solid, we have that

$$\ln \Omega(n, N) = \ln [(n+N)!] - \ln [N!] = \ln [n!] \quad (23)$$

$$\approx (n+N) \ln(n+N) - N \ln N - n \ln n \quad (24)$$

In the high temperature limit, for $n \gg N$, then

$$\ln(n+N) = \ln \left(n \left(1 + \frac{N}{n} \right) \right) \quad (25)$$

$$= \ln n + \ln \left(1 + \frac{N}{n} \right) \quad (26)$$

$$\approx \ln n + \frac{N}{n} \quad (27)$$

Therefore,

$$\ln \Omega(n, N) \approx N \ln \left(\frac{n}{N} \right) + N \quad (28)$$

Therefore,

$$\Omega = e^{\ln \Omega} \approx e^{N \ln(n/N) + N} \quad (29)$$

$$= \left(e \frac{n}{N} \right)^N \quad (30)$$

$$= \left(\frac{eU}{N\hbar\omega} \right)^N \quad (31)$$

Here, $n = U/\hbar\omega$. Therefore,

$$\Omega \sim U^N \quad (32)$$

For a lot of systems, we have that

$$\Omega \sim U^f \quad (33)$$

In an Einstein solid, $f \sim N$.

Entropy

Here, we define Entropy as:

$$\sigma = \ln \Omega \quad (34)$$

This is so that σ is additive when you add two systems together and calculate its multiplied multiplicity. Then,

$$\sigma(U, N) = N \ln \left(\frac{eU}{N\hbar\omega} \right) \quad (35)$$

If you double the system, then $U \rightarrow 2U$ and $N \rightarrow 2N$, then $\sigma \rightarrow 2\sigma$. Note, this is only true in the high temperature limit.

Paramagnetic

Collection of non-interacting spins. Then,

$$N = n_{\uparrow} + n_{\downarrow}$$

Such spins have a dipole moment

$$m\vec{u} = g\frac{q}{2m}\vec{s} \quad (36)$$

Then, the energy is then: (for a B field in the z direction)

$$U = -\vec{\mu} \cdot \vec{B} = -\mu_z B \quad (37)$$

Such that

$$\mu_z = g\frac{q\hbar}{2m}m_z, \quad m_z = -s, -s+1, \dots, s-1, s \quad (38)$$

Here, $s = 1/2$ (i.e., we only deal with particles with a spin-1/2.) There are only two energy levels. Working within the binary system for $s = 1/2$, then

$$\Omega(n_{\uparrow}, N) = \frac{N!}{(n_{\uparrow}!)(N - n_{\uparrow})!} \quad (39)$$

Then

$$\Omega(U, N) = \frac{N!}{(N/2 - U/(2\mu B))!(N/2 + U/(2\mu B))!} \quad (40)$$

Here, we used the fact that $U = n_{\uparrow}(-\mu B) + n_{\downarrow}(\mu B)$. Here,

$$n_{\uparrow} = \frac{N}{2} - \frac{U}{2\mu B} \quad (41)$$

The above equation is necessary for the homework. We can also then use Stirling's approximation to then obtain

$$\ln \Omega(n_{\uparrow}, N) \iff \sigma(n_{\uparrow}, N) = N \ln N - (N - n_{\uparrow}) \ln(N - n_{\uparrow}) - n_{\uparrow} \ln n_{\uparrow} \quad (42)$$

You can also re-express this using equation (40) to get $\sigma(U, N)$. $\sigma(U, N)$ as a function of U looks like a half parabola with a maximum at $U = 0$.